Lab 9: OSM and Interactive Web Maps

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Urban Informatics

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Reminder: Final Project Groups

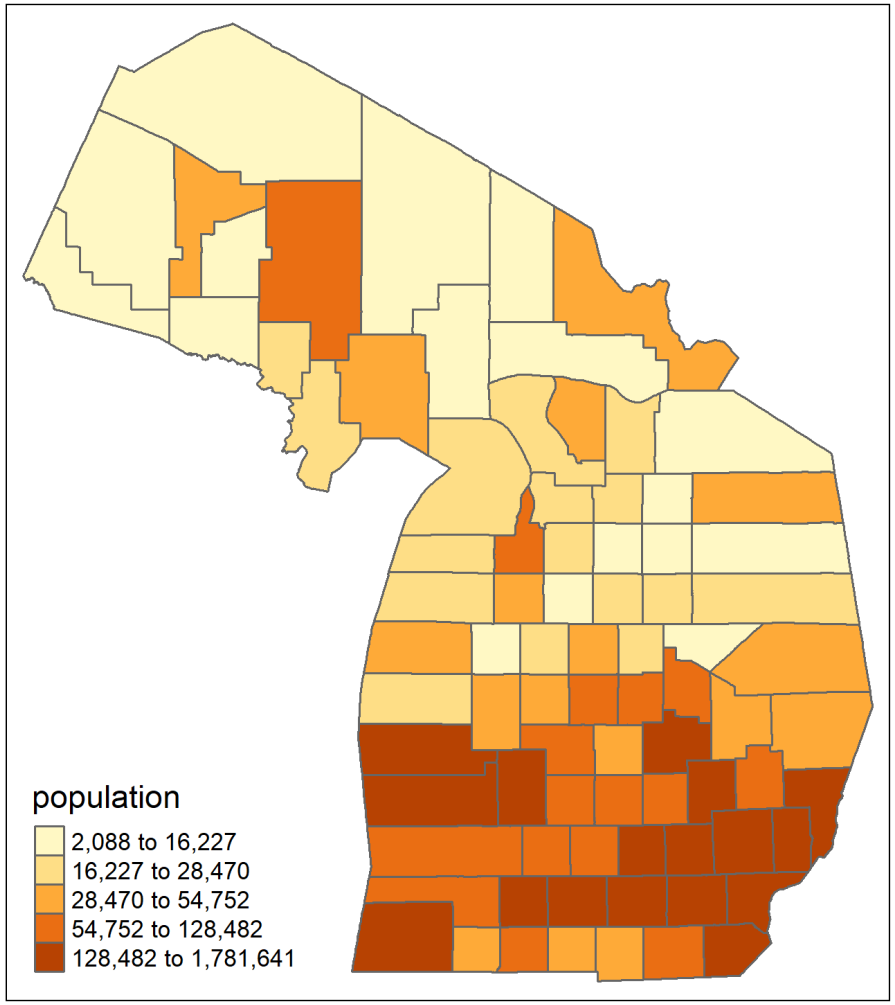
Follow this link

to rank your preferred topics / groups for the final assignment.

Static Map vs. Interactive Web Map

So far we have mostly used **static maps:**

- A fixed image of a map (e.g., PNG, JPEG, PDF)
- Created with *geopandas* / *matplotlib* (Python), *tmap* (R), or GIS software
- Not interactive, i.e., users cannot pan, zoom, or click figures.



Static Map vs. **Interactive Web Map**

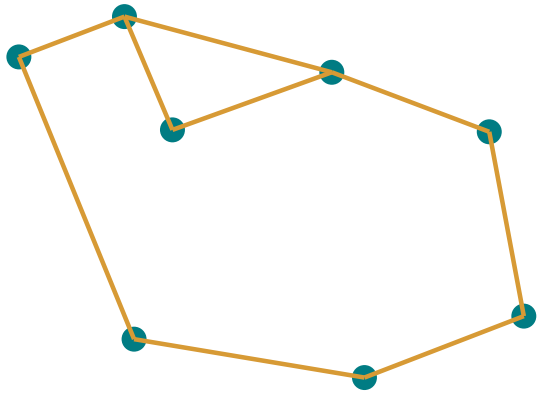
However, this lab focuses on **interactive <u>web</u> maps**

- Can change based on input (e.g., filtering POIs by type, turning layers on or off, showing popups)
- Requires HTML, CSS, and Javascript (folium + leaflet)
- Can be hosted online to share with other users.



Interactive Maps with OSM

Open Street Map



OSM is an online database of mapped urban features (e.g., roads, POIs)

OSMnx / Osmdata

```
bbox = (west, south,
        east, north)

ox.features_from_bbox(
    bbox,
    tags={'building': True}
)
```

We use OSMnx (Python) or Osmdata (R) packages to access it

GeoDataFrames / SF

feature	geometry
building	polygon
road	line
amenity	point

And store its input in datasets with geographic attributes

Interactive Maps with OSM

feature	geometry
building	polygon
road	line
amenity	point

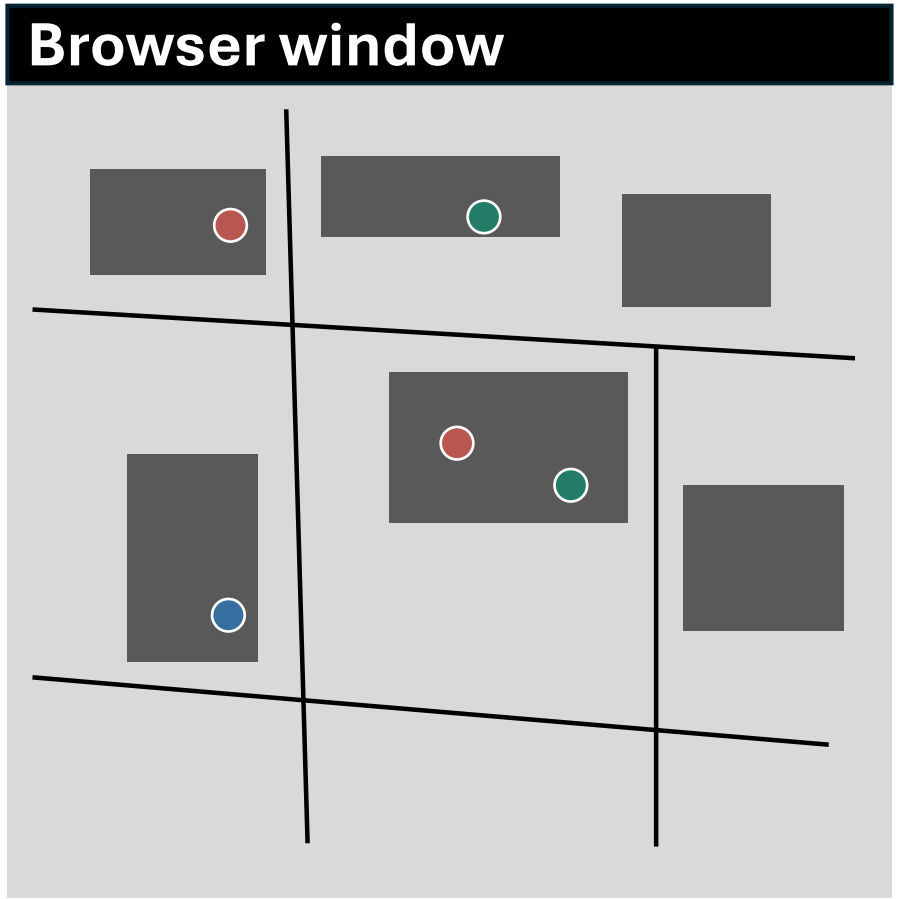
Geopandas and folium work to translate code (e.g., see below) into HTML, CSS, and JavaScript

```
# Building layer
m = building.explore(...)

# Add road layer
roads.explore(m=m, ...)

# Add amenity layer
amenity_point.explore(m=m, ...)

# Save interactive file
m.save("lab9_map.html")
```



Web Hosting: Shareable Map Links

Local Rendering:
For a simple application, you can write HTML, CSS, JavaScript in one .html file and open it locally in your browser. This is the same type of file you get if you export a HTML file from Python/R.

Web Hosting:
If you want a sharable, public link on the web, you need to **host** your .html file on a server. In this lab we use **GitHub Pages** but there are other options (e.g., RPub for R).

What is GitHub and GitHub Pages

GitHub is an online platform for storing projects, tracking changes over time, and collaborating with others.

GitHub Pages is a free GitHub service that takes files stored on GitHub and publishes them as a public website. Most other hosting service costs money (e.g., you have to buy a domain name).

Example: **Professor Liang** uses GitHub pages to host her personal website and all interactive maps. Feel free to explore her repository here: https://github.com/xiaofanliang?tab=repositories

Publish a HTML Map on GitHub

- You create a map in Python or R.
- You save/name it as a html file.
- You upload it to GitHub.
- You click “Enable Pages.”
- GitHub gives you a public web address, e.g.:
  `https://<your_username>.github.io/REPO/<map_name>.html`

How to write HTML, CSS, JS in Python/R for Maps? → Leaflet.js

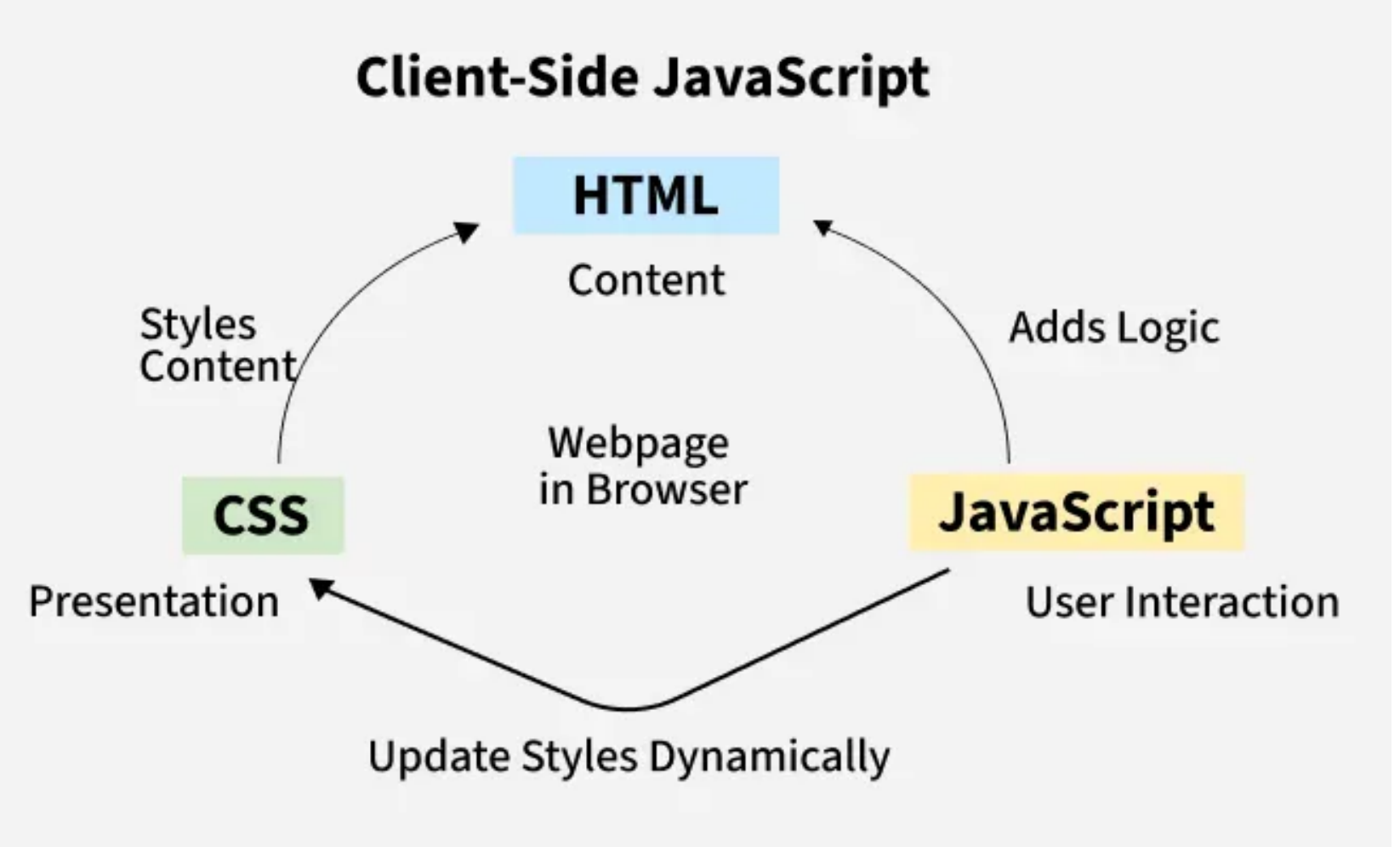
- Leaflet.js is an **open-source** JavaScript library for interactive webmaps.
- You don't need to know how to write HTML, CSS, JS.
  - The ***folium*** package in Python and ***leaflet*** package in R translate R/Python commands into instructions that can execute in your browser.
  - They take in Python or R code and create an interactive map you can export a HTML file you can host somewhere (e.g., GitHub Pages)
- Still: knowing basic HTML, CSS, and Javascript can help with customization, understanding your exported HTML, debugging, etc.

Basic Web Concepts

HTML (HyperText Markup Language) defines the structure of a webpage (e.g., headings, paragraphs, maps)

CSS (Cascading Style Sheets) controls the styling of a webpage (e.g., colors, fonts, layout)

JavaScript adds interactivity to a webpage (e.g., buttons, dynamic maps, animations)



https://www.geeksforgeeks.org/javascript/javascript-tutorial/

HTML Raw Code (L) & Web Rendering (R)

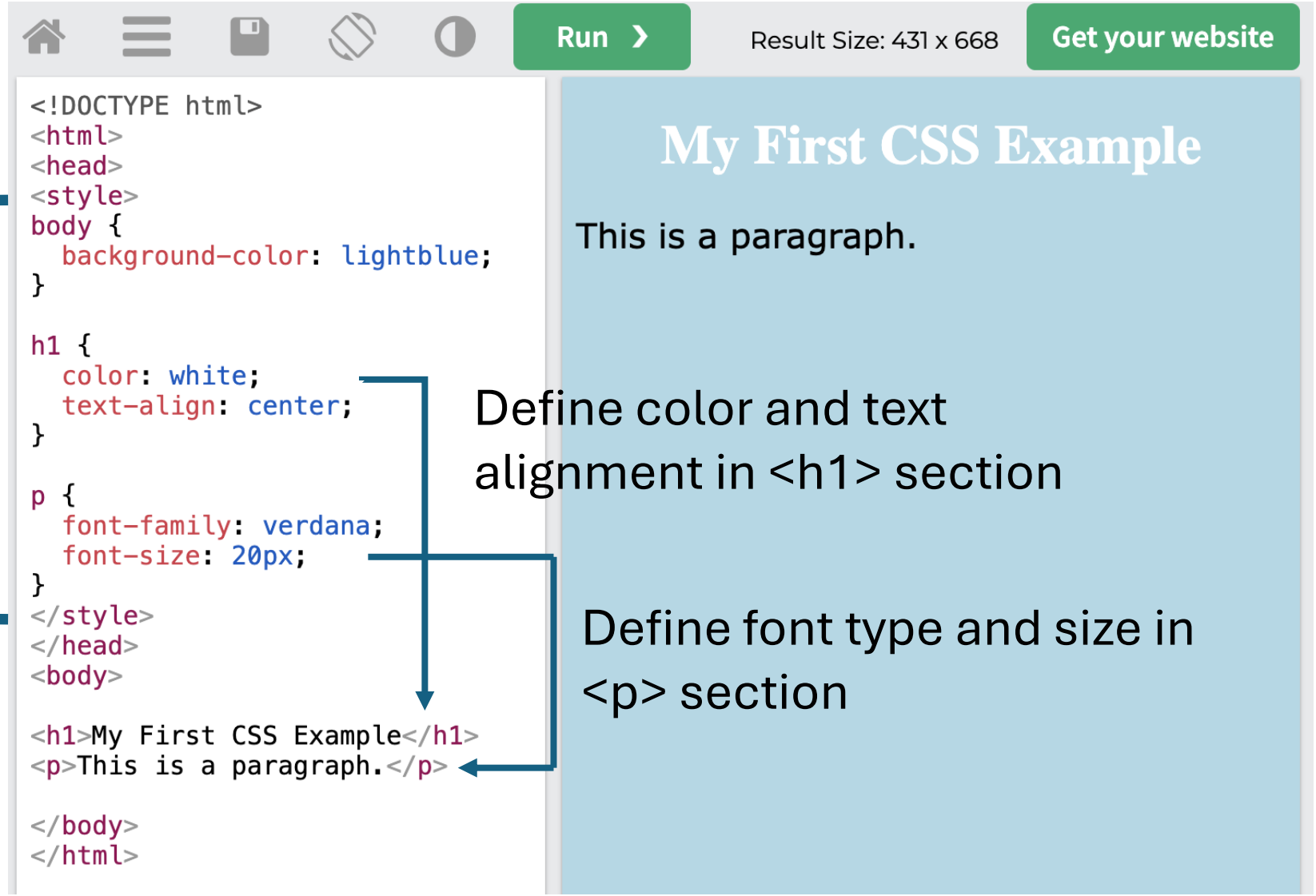


https://www.w3schools.com/html/tryit.asp?filename=tryhtml_intro

CSS Raw Code (L) & Web Rendering (R)

CSS rules are wrapped in <style> tags nested in a **HTML** file

HTML content



Define color and text alignment in <h1> section

Define font type and size in <p> section

https://www.w3schools.com/css/tryit.asp?filename=trycss_intro

JavaScript Code (L) & Web Rendering (R)

Run ›

Result Size: 681 x

```
<!DOCTYPE html>
<html>
<body>
<h1>My First JavaScript</h1>

<button type="button"
onclick="document.getElementById('demo')
.innerHTML = Date()">
Click me to display Date and Time.
</button>

<p id="demo"></p>

</body>
</html>
```

id = ‘demo’

My First JavaScript

Result of `Date()`

Click me to display Date and Time.

Mon Mar 02 2026 09:17:11 GMT-0500 (Eastern Standard Time)

When clicked, the button runs JavaScript code that finds the element wih `id = ‘demo’` and inserts the current date and time (from `Date()`) inside.

Resources for Learning More

- OSMnx docs: essential urban mapping package/API for OSM
- MDN Web Docs: web development documentation
- Leaflet : Official Leaflet JS tutorials and API
- Python Folium : Python package documentation
- R Leaflet : R package documentation
- W3Schools: Interactive tutorials for HTML/CSS/JavaScript

Next

- **Review explanations and practice examples** -> do practice questions -> complete the quiz
- Canvas -> Assignment -> Lab 9 quiz **(this is different from others)**
  - **Submit by next lab**
  - If R, submit your R code (if any) in ***.R*** format **for the Quiz questions ONLY**
- Feel free to leave when you are done with lab